

Laser's Phase Influence on the Polarization

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Description of the effect

It turned out that the polarization measured may depend on the slit-attenuator configuration in the injector. HAPPEX was running with the slit retracted (completely open) and no laser attenuation while the Moller measurements typically used configurations with the slit tight and no/low attenuation. The latter was done in order to minimize cross-talks from the lasers of Halls B,C running at a phase opposite to Hall A laser. Additional important parameter of the polarized gun is "phase". With a narrow slit the "phase" defines what time phase of the laser RF pulse comes through the slit, which may vary from the maximum of the pulse to a tail, while with the slit retracted practically all the pulse passes through whatever is the "phase" parameter. The "phase" was readjusted 2-3 times per week.

The influence of the phase adjustment on the polarization was measured at the end of May at the conditions as follows:

- Moller target angle was 160° . No measurement at 20° was done and therefore the polarization measured was good to a certain multiplicative factor.
- For a given combination of attenuation, slit and phase parameters. 6 Moller runs were taken, for the polarization and the beam rate the average values were calculated.
- For the measure of the beam intensity the 2-arm coincidences of the Moller setup were used.
- The cross-talk from the laser B and C was measured by turning the laser A off.

The results are presented on a plot which shows:

1. The Moller coincidence rates depending on the phase, for the slit parameters of 16.4 and 17.3. At 16.4 the measurements were done at attenuations of 500 and 300, but the results were all normalized to 300, while at 17.3 the measurements were done at attenuation 250. The arrow indicate the regular phase of that period. Apparently, the regular position is on a tail of the distribution. For the more narrow slit of 16.4 the shape of the peak is sharper than for the wider 17.3 slit. As a matter of fact the regular slit parameter for the Moller measurements was about 15.6.
2. The relative cross-talk from the lasers B and C depends on the phase. The results were corrected for this cross-talk.
3. The beam polarization is flat on the peak top but drops by about 10% at the tail of the peak. Therefore, if during the Moller measurements the phase is adjusted to the tail of the peak, the results should be systematically lower than the average for the whole distribution.
4. The beam current in the injector was measured the next day. For some reason the peak became broader and a phase shift occurs with respect to the previous day. This day, the Moller measurement would not be distorted, since the default phase sits at the top of the peak.
5. This plot shows the beam current, measured on the injector, depending on the slit parameter. The phase was at -14° - at the top of the beam current profile. The open and filled circles present the data taken at

the laser attenuation of 500 and 600. The data were approximated with a polynomial with the coefficients as follows: 0.32209,-0.01468,0.29857,0.21897,-0.09835.

6. This plot shows the relative beam current, measured on the injector, depending on the laser attenuation. Since the measurements were done at different slit sizes, the results were normalized to a certain slit size using the polynomial defined in 5). The data can be approximated with a curve $a \cdot \sin((x - x_0) \cdot \pi / 550)^2$, with $x_0 = 0$. (This is different from $x_0 = 50$, mentioned once by the accelerator people. I have not found a reference to their measurements).

We have not calibrated the beam current dependence on the laser power parameter.

Correction of the results already taken

For all the Moller measurements done so far no record of the phase have been taken. Furthermore, if it had been taken it would be of little use since the peak position is not fixed to a certain phase, as follows from the plots 1) and 4). For each measurement something like the curve 1) or 4) should be measured and the phase should be adjusted to the top of the distribution.

However, since the beam current, laser power, attenuation and slit parameters have been recorded, one can reconstruct the final beam attenuation. Typically, for all the measurements the beam current, the laser power and the laser attenuation were the same. The beam current drops sharply at the edges (see plots 1) and 4)) by a factor of about 5-10, and should drop even sharper if a narrower slit is used, which was the case for the Moller measurements. Therefore, it is unlikely that the phase was accidentally chosen somewhere on the slope, but rather it was either on the top or on the tail. The polarization suffers a 4% drop when the beam current drops by a factor of 5. For a given slit parameter one can decide whether it was the top or the tail using the plot 5).

In order to filter the measurements done for HAPPEX, looking for those likely been done on the tail, we used 3 values, stored for every run: the beam current, the laser attenuation and the slit parameter. We had no calibration of the beam current dependence on the laser power and therefore could not take it into account. Fortunately, for the most of the HAPPEX measurements the laser power stayed the same, Assuming that no other parameter but the laser attenuation and the slit could change the beam current, we found a factor of the "additional" beam attenuation, attributed to the phase position off the peak, normalizing to a run definitely taken on the peak. The definite "tail" run had this factor of about 0.2, consistent with the the plot 1). This "attenuation factor" is used to select the measurements taken on the tail. The best we can do is to reject these measurements.

Summary

A potential source of systematic errors was found, associated with the fact that Moller polarimeter typically runs at a beam current of about 1% of the current used by the experiments. There are two reasonable ways to reduce the current to such a low value:

1. Attenuate the laser beam
2. Attenuate the electron beam using a slit with a variable width.

The 1-st method is not acceptable if the other halls are running because of a considerable cross-talk from the dark currents of the other lasers, typically polarized in a direction, opposite to the laser of Hall A. At no attenuation the effect of the cross-talk is below 0.1%, but may reach a few % at a strong attenuation. The second method looks preferable, but one has to be sure that the narrow slit selects the plateau of the laser RF structure, but not a tail where the beam polarization is lower by 4-10%. The "phase" parameter of the injector setup is used for appropriate adjustments.

Concerning the measurements already taken, for the HAPPEX run of Apr-May 1999 one is able to reconstruct whether the measurement was taken on the plateau or on the tail. For earlier runs it is not

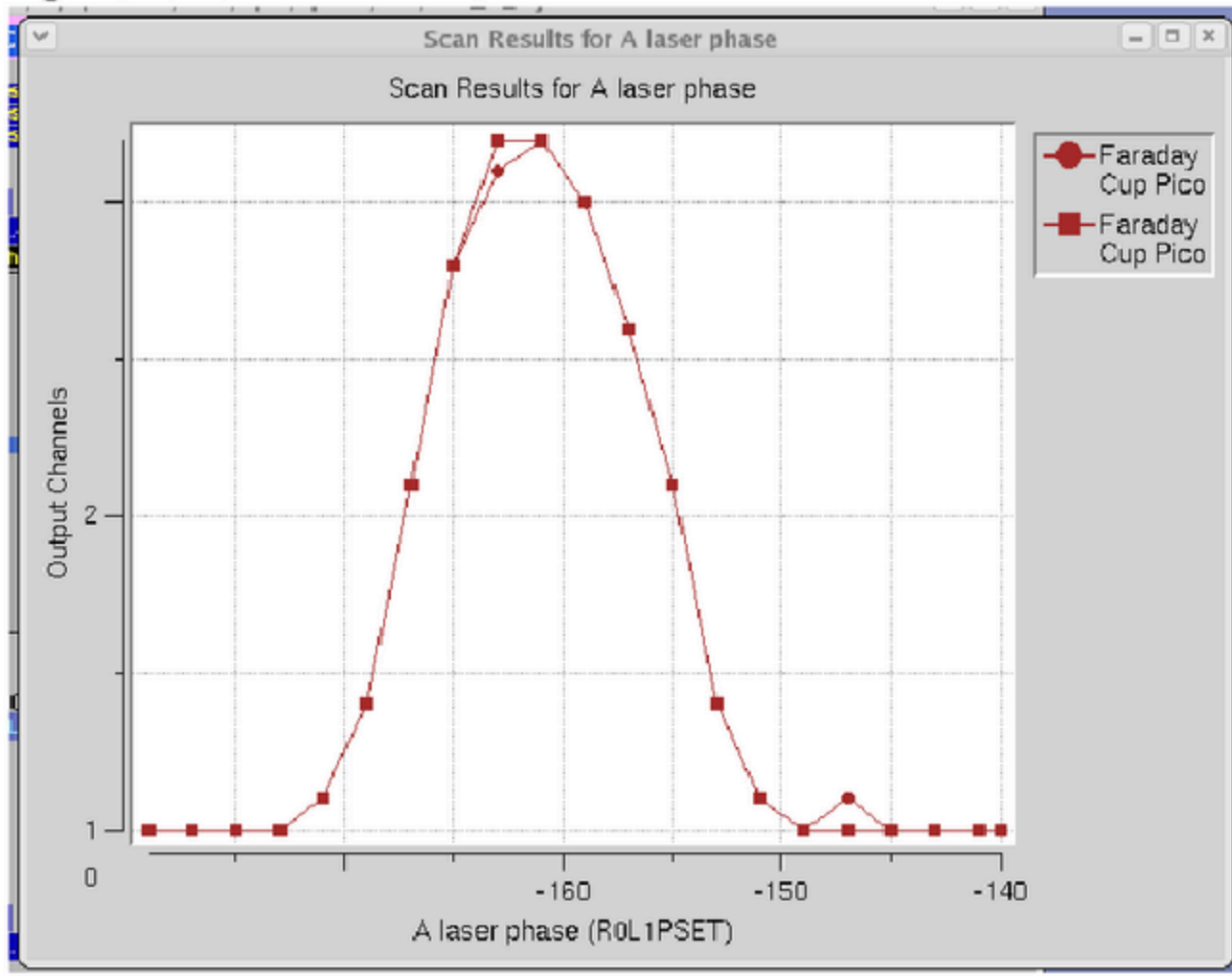
possible. However, it is just for that HAPPEX period the injector might be tuned in a way that the "phase" was off the plateau, since the experiment demanded as high beam current as possible and an asymmetric phase tune increased the current by 1-2uA, or so, incorporating the late in time tail of the RF laser structure.

More details

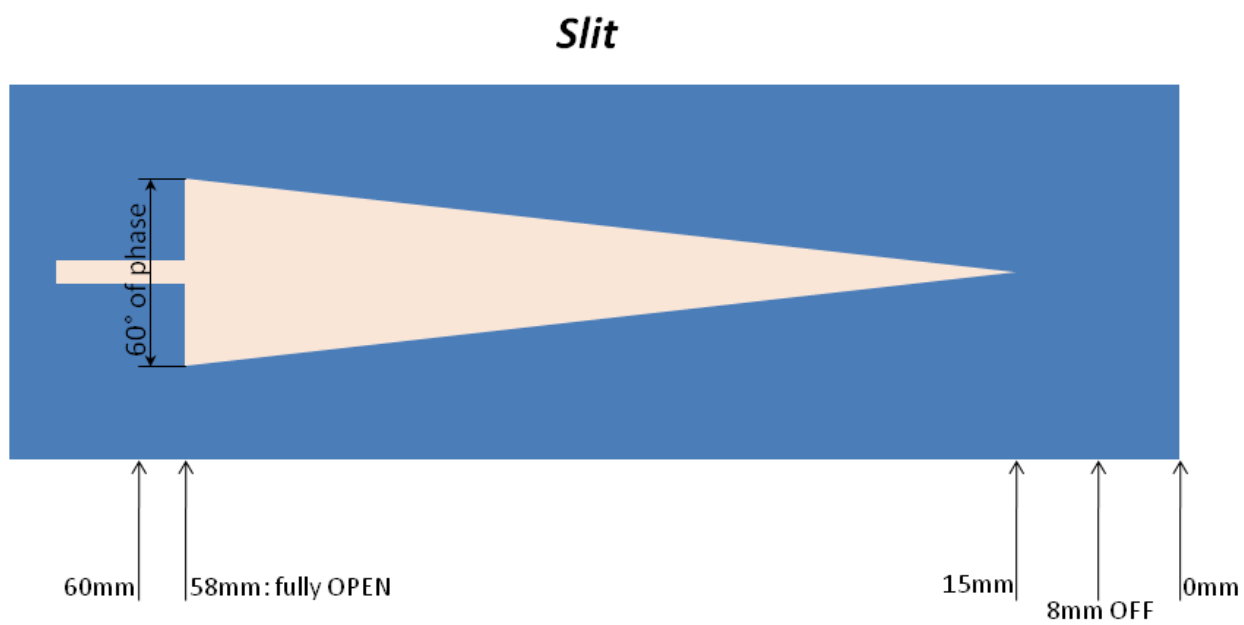
There is an additional information about the injector and laser phase.

In 1999 strained GaAs photo-cathode was used in the injector. In 2009 super-lattice photo-cathode is used in the injector. In the super-lattice photo-cathode electrons are produced in more thin layer and phase profile is different.

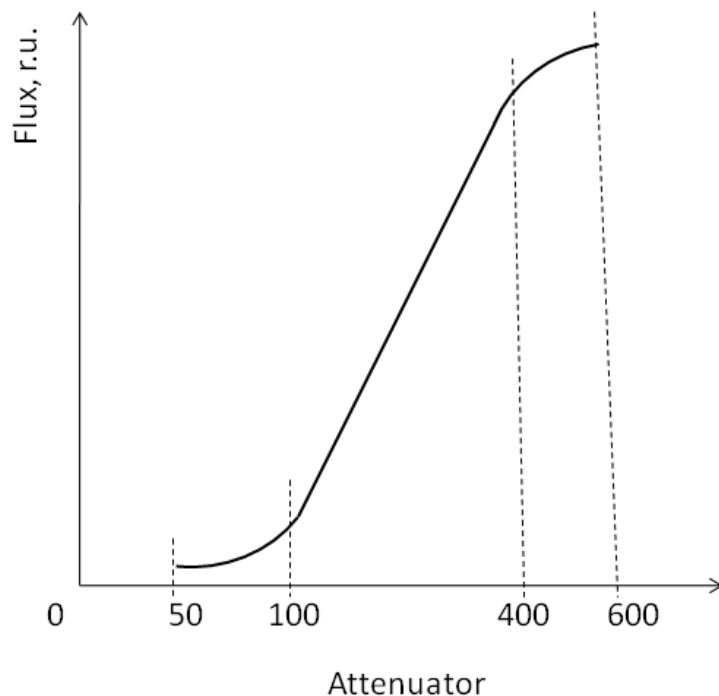
Fig. 1 [10/02/2009 09:56:56]



Injector slit sketch is presented on the plot below. Angle 60° is Hall A laser phase range for fully open slit.



Profile of beam current vs laser attenuator dependence is shown on the plot below:



For the laser phase study with Moller polarimeter it is the best to have MCC turn OFF the prebuncher, which can push the head/tail of the bunch to different temporal locations. Unfortunately, it will affect other Halls beam condition. Laser phase study on the injector is free of this effect.

2009 Results

In October 2009 new laser phase study with supper-lattice photocathode was done.

Hall A laser phase setting for HAPPEX-III running and for the Moller measurements was the same.

All measuremens were done with the prebuncher turned ON.

No cross-talk from the lasers B and C was found.

Preliminary results are presented on a plot 1 and on a plot 2 which shows:

- The Moller coincidence rates depending on the phase for the slit parameters of 14.6, 15.45 and 23.7.
- The beam current in Hall A for the slit parameters of 14.6, 15.45 and 23.7. For safety reasons at the beam current higher than 1uAmp the Moller target was moved out of the beam. Moller coincidence rate and beam polarization was not measured at high current. Different shape and shift of curves at slit parameter 15 and 23 can be explained (by M. Poelker) with the prebuncher effect.
- The beam polarization dependence of laser phase. There are a few points (174° , 163.8° , 148° and 144°) with large statistics. These points give the same polarization in wide laser phase range.
- This plot shows Moller results for different combinations of slit and attenuator. Every color is corresponded to the beam polarization was measured with Moller polarimeter at the same day but with

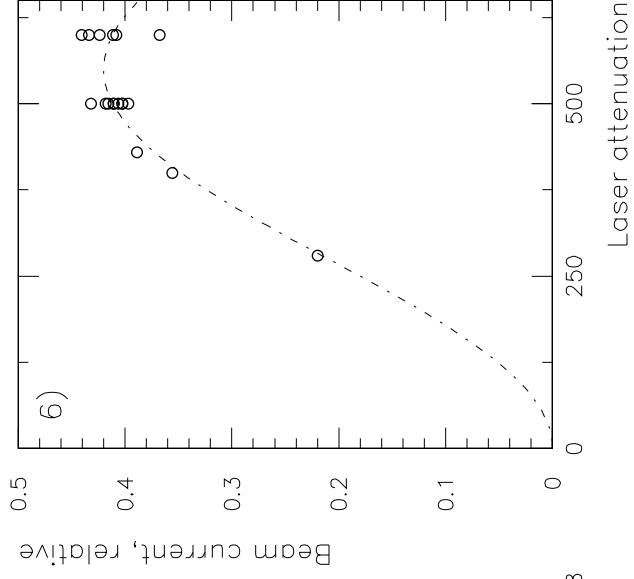
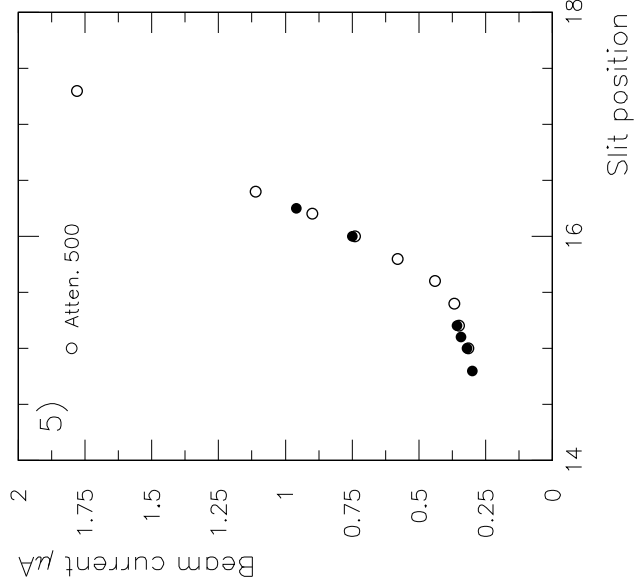
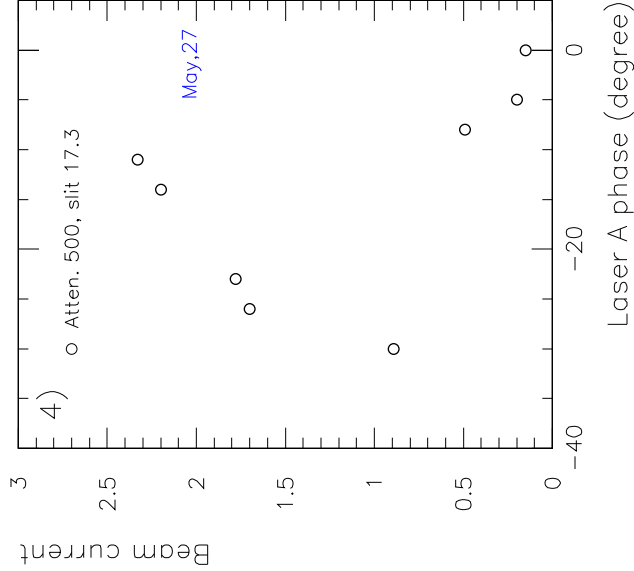
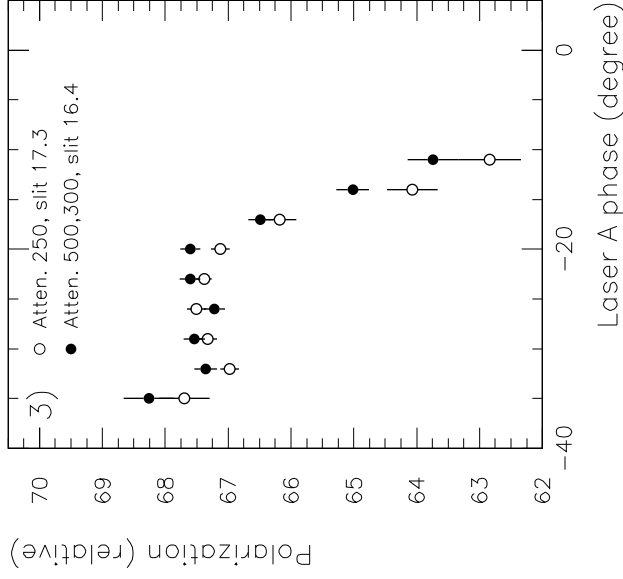
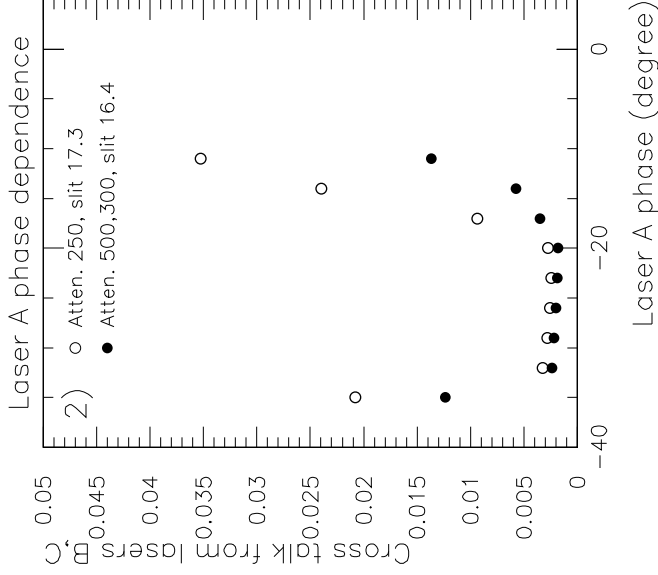
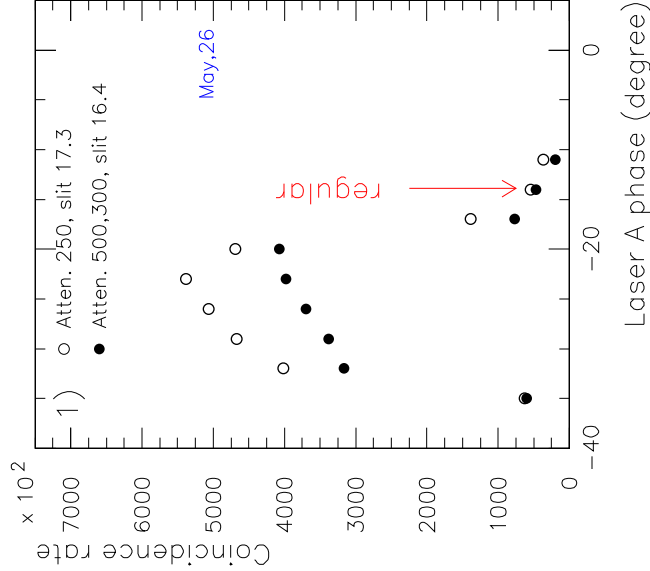
different slit/attenuator settings. It is seen that for the laser phase 163.8° Moller results stay the same in wide range of slit/attenuator settings.

Dependence of the beam current of slit parameter at the same laser phase and attenuator is shown on a plot. Dependence of the beam polarization of combination of laser attenuator and slit parameters is shown on a plot.

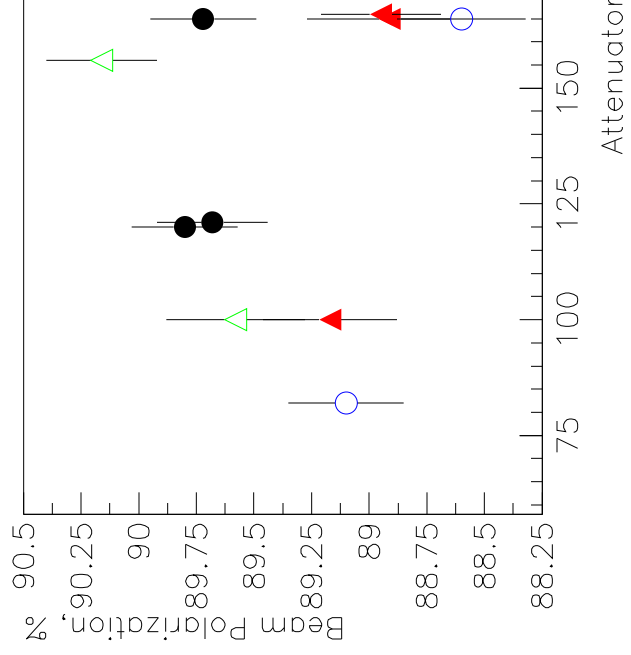
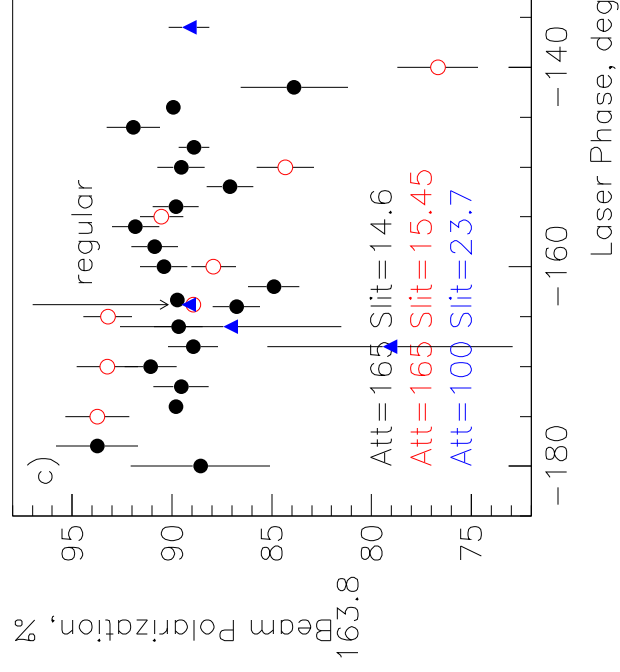
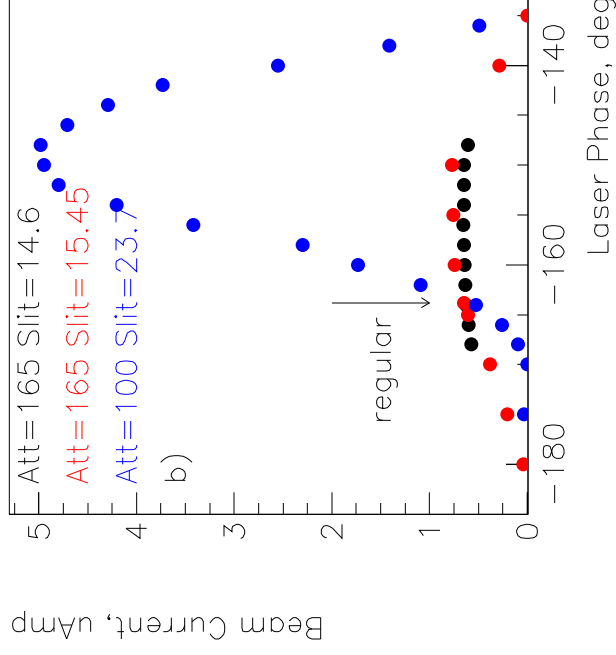
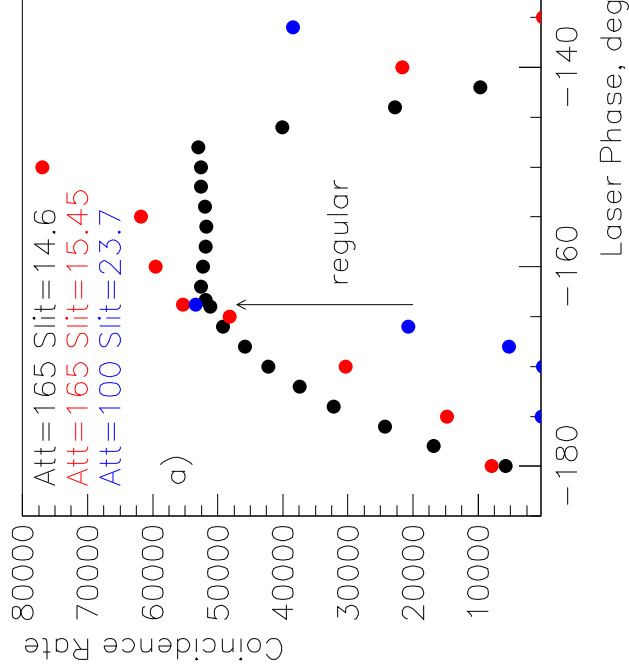
To do list

In order to finish Hall A laser phase study the next measurements should be done:

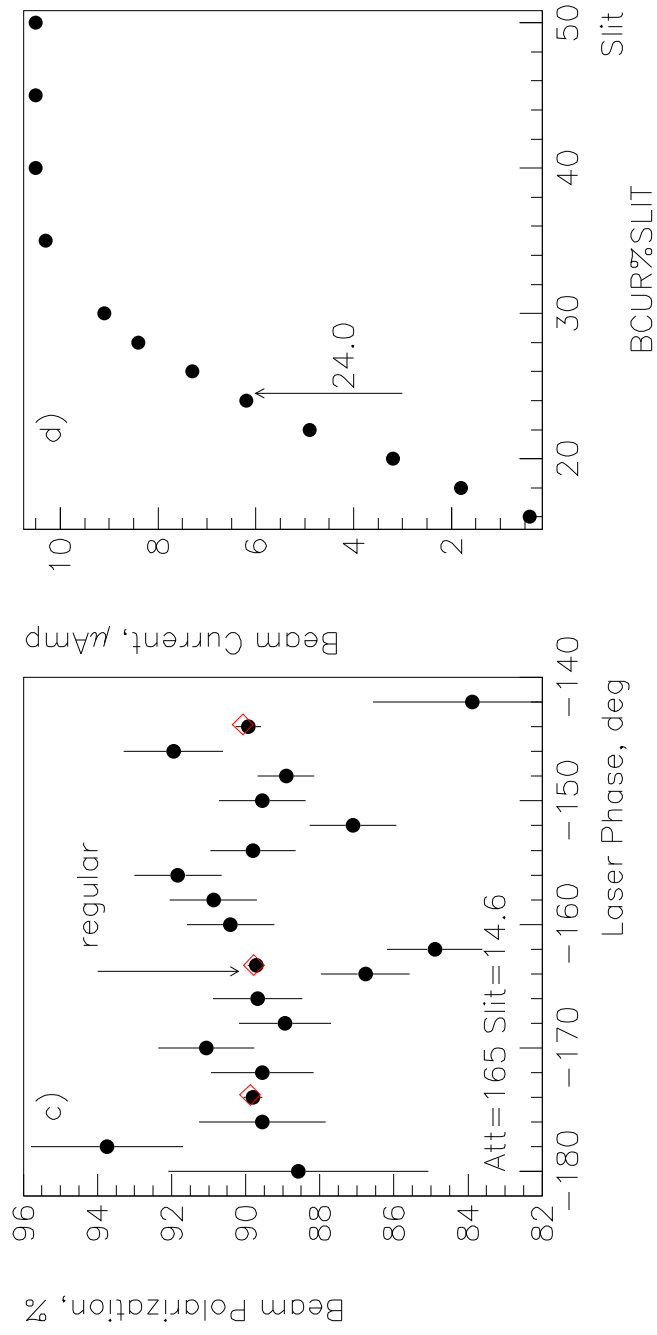
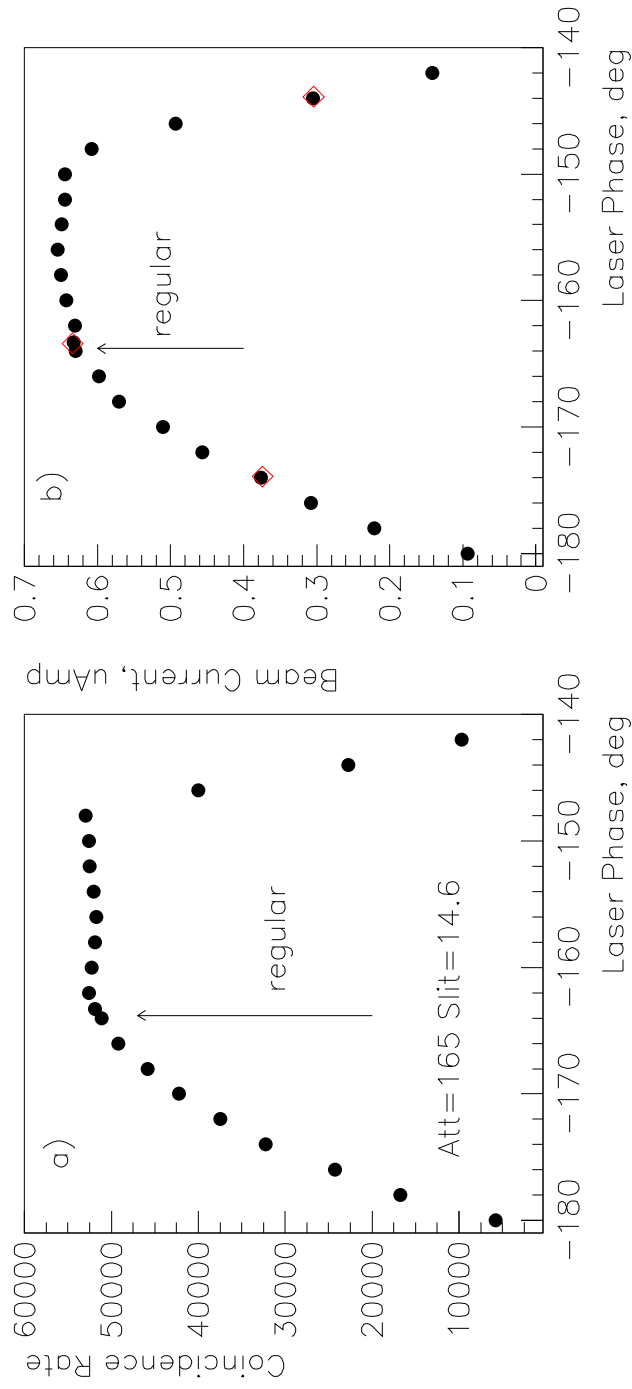
1. Hall A laser phase scan (beam current measurement) with open slit ($\sim 60\text{mm}$) and attenuator 100, 150 and 200 (on the injector with the prebuncher turned OFF).
2. Measurement of the beam current depending of attenuator parameter at the same laser phase and slit (on the injector with the prebuncher turned OFF).
3. Set 100uAmp of the beam current in Hall A. Adjust the beam current to 0.6uAmp with attenuator. Measurement of the beam polarization for different laser phase (from -180° to -140° with step 5°) with Moller polarimeter (prebuncher ON).

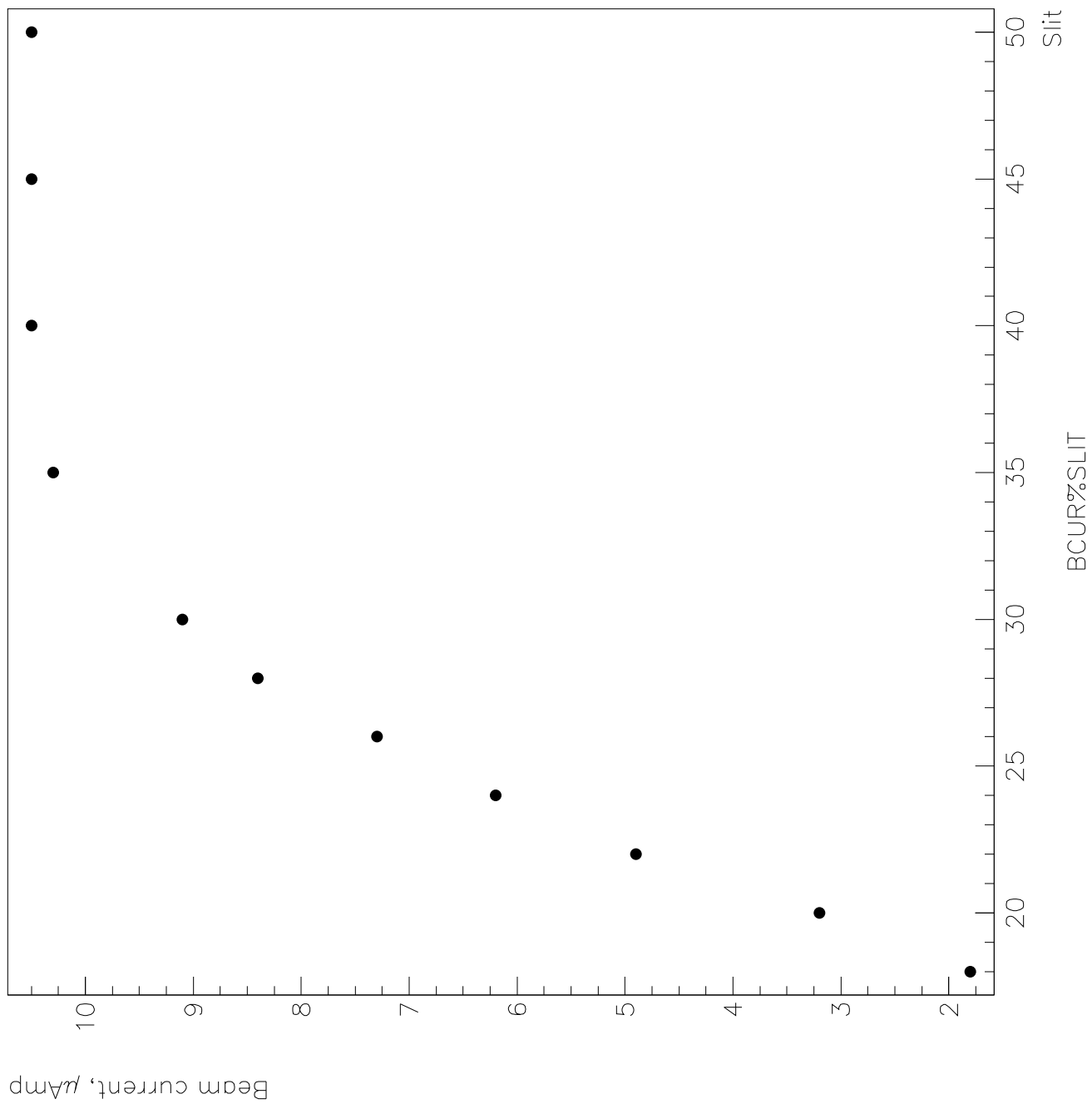


Laser phase dependence



Laser phase dependence





Laser Phase Dependence

Slit vs. Attenuator

